

Executive Summary

Salifort Motors — Employee Retention Prediction | Google Advanced Data Analytics Capstone

Author	Dataset	Rows / Columns	Champion Model
Ahmad Daniel	HR_capstone_dataset.csv	11,991 / 10	Random Forest (GridSearchCV)

ISSUE / PROBLEM

Salifort Motors, a global alternative energy vehicle manufacturer (100,000+ employees), is experiencing a high rate of employee turnover. Senior leadership is concerned about the financial and operational costs of attrition. Finding, interviewing, and training new employees is time-consuming and expensive. The HR department surveyed a sample of employees to understand what might be driving departure. The core business question: **What factors make an employee likely to leave the company — and can we predict who will leave before they do?**

RESPONSE

Data Preparation

- Dataset: HR_capstone_dataset.csv (14,999 rows, 10 columns)
- Renamed columns for consistency: Work_accident, average_monthly_hours (typo fixed), time_spend_company (renamed to tenure), Department
- No missing values found
- Dropped 3,008 duplicate rows (~20% of data)
- Identified outliers in tenure via IQR method (removed for logistic regression only)

EDA Key Findings

- All employees with 7 projects left the company
- Employees working 240-315 hrs/month had near-zero satisfaction
- 4-year tenured employees show an unusually low satisfaction level
- Employees with 6+ years at the company overwhelmingly stayed
- Very few employees who were promoted in the last 5 years left
- No single department showed dramatically different attrition rates

Modeling Approach

- **Approach A — Logistic Regression:** salary ordinal-encoded, department dummy-encoded, tenure outliers removed, 75/25 train/test split (stratified, random_state=42)
- **Approach B — Tree-based Models:** Decision Tree + Random Forest with GridSearchCV (cv=4, refit=roc_auc), 75/25 split (stratified, random_state=0)

Feature Engineering (Round 2)

- Dropped satisfaction_level (potential data leakage)
- Created binary overworked feature: 1 if avg_monthly_hours > 175, else 0
- Dropped average_monthly_hours (replaced by overworked)

Evaluation Metric

- Primary: AUC (Area Under ROC Curve) for tree-based models
- Secondary: Precision, Recall, F1-score, Accuracy
- Champion: Random Forest Round 2 (feature-engineered, leak-safe)

IMPACT

The Random Forest model (Round 2, with feature engineering) is the recommended champion model. It achieves near-perfect predictive performance while mitigating data leakage concerns by relying only on features that would realistically be available prior to an employee's departure decision.

Model	Precision	Recall	F1-Score	Accuracy	AUC
Logistic Regression	~80%	~83%	~80%	~83%	N/A
Decision Tree (Round 1)	~87%	~91%	~89%	~96%	~97%

Random Forest (Round 1)	~88%	~90%	~89%	~96%	~98%
Decision Tree (Round 2)	~85%	~90%	~87%	~95%	~94%
Random Forest (Round 2) ★	~87%	~90%	~88%	~96%	~96%

★ Champion model — recommended for production use.

KEY INSIGHTS

Top Predictors of Employee Departure

Both the Decision Tree and Random Forest models consistently identified the same top four features driving employee attrition:

#1 last_evaluation	Most important feature — high performers who are overworked and unrecognized are most at risk.
#2 number_project	Employees with 7 projects had a 100% departure rate. Optimal range is 3-4 projects.
#3 tenure	Four-year employees show unusually high dissatisfaction. Employees with 6+ years tend to stay.
#4 overworked	Working >175 hrs/month is a strong departure signal, especially without promotion.

Recommendations for Stakeholders

- **Cap projects at 5.** Assign 7 projects to no one — 100% of these employees left.
- **Investigate 4-year milestone.** Conduct targeted interviews with 4-year employees; consider promotion criteria review.
- **Reward long hours or reduce them.** Employees should not be expected to work 200+ hours/month without recognition or additional compensation.
- **Clarify overtime and time-off policies.** Ensure all employees are informed of their rights and entitlements.
- **Reform evaluation scoring.** Proportionate effort-based recognition — not hours-only scoring.
- **Culture review.** Hold department-level and company-wide discussions to surface systemic dissatisfaction drivers.

Next Steps

- Explore model performance without last_evaluation (possible leakage / infrequent surveys)
- Perform K-means clustering to segment employee risk profiles
- Retrain model as new HR survey data becomes available
- Ensure model predictions are used to improve working conditions — not to penalize employees or enable discriminatory action.